Reinforcement Learning - Evaluation Report

Focus: Physical Placement Quality and Routability Validation

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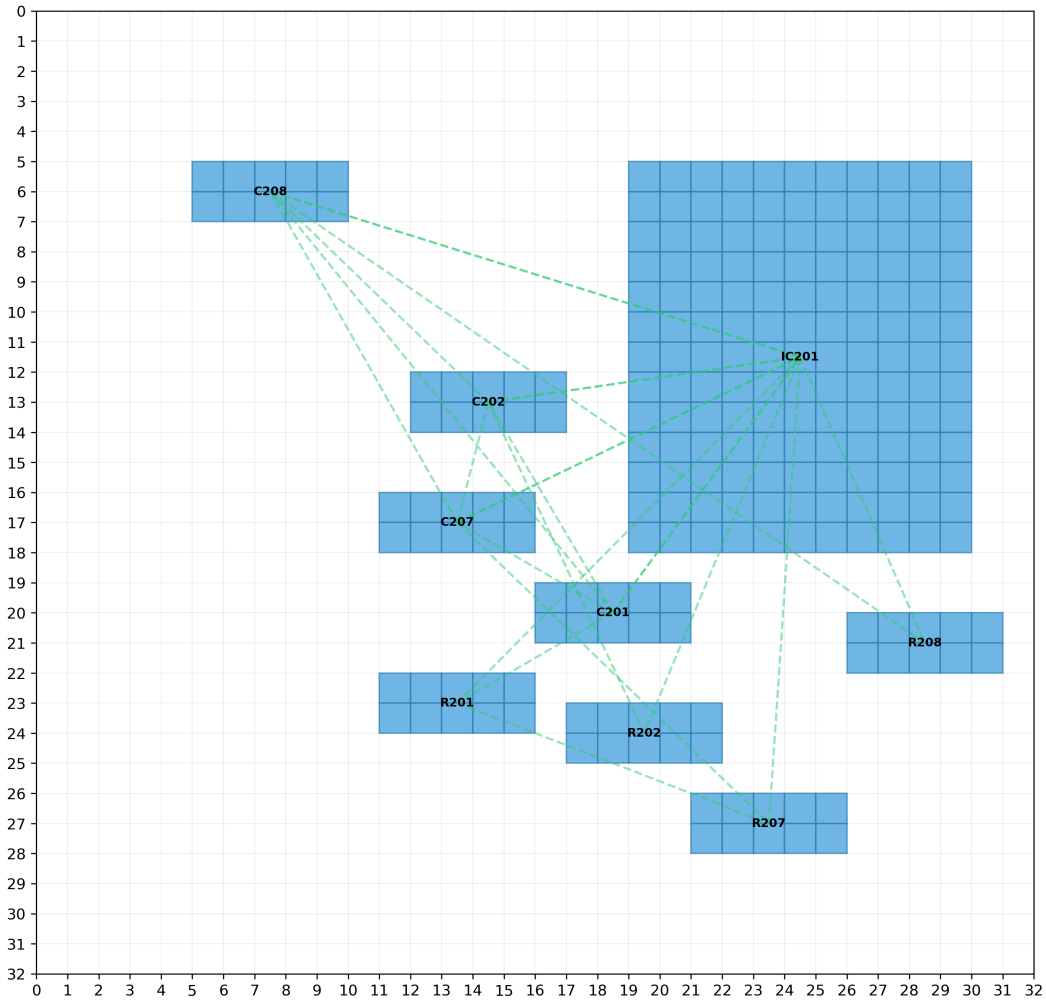
Validation Environment

Hardware & Software	
Platform	Linux-6.17.0-20-generic-x86_64-with-glibc2.39
Python	3.12.3
PyTorch	2.11.0+cu130

Placement Visualization

Physical PCB Placement & Routing Skeleton



Physical Routing & Placement Metrics

These metrics represent the 'ground truth' for layout quality.

Metric Name	Value	Description
Hpwl	152.5000	Half-Perimeter Wire Length (Estimated)
Normalized Hpwl	0.2648	Physical metric
Drc Pass Rate	1.0000	Physical metric
Overlap Rate	0.0000	Physical metric
Routability Proxy	0.0411	Heuristic estimate of routing success
Total Nets	6	Total number of connections in netlist
Components Placed	9	Count of successfully placed components
Total Components	9	Physical metric
General Routes Established	3	Physical metric
Differential Routes Established	3	Physical metric
Routed Wirelength Mm	54.8877	Actual Routed Wire Length from Rust Router
Num Vias	0.0000	Number of layer transitions (vias)
Num Bends	12.0000	Physical metric

Final Routing Status: PARTIAL